Foundation Model Adaptation for Meningioma Growth Forecasting from Multi-Modal 3D MRI

A Gaussian Process Hierarchy over Disentangled Latent Trajectories

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Abstract

Meningiomas are the most prevalent primary intracranial tumours, yet predicting their growth trajectories from longitudinal MRI remains an open challenge due to heterogeneous growth dynamics and limited longitudinal data. We present a four-phase pipeline that adapts BrainSegFounder—a SwinUNETR foundation model pre-trained on 41,400+ brain MRI subjects—to the meningioma domain. Phase 1 applies Low-Rank Adaptation (LoRA) to the encoder's Stages 3–4 using segmentation as a proxy task on BraTS-MEN 2024 (1,000 subjects). Phase 2 trains a Supervised Disentangled Projection (SDP) that maps frozen encoder features $\mathbf{h} \in \mathbb{R}^{768}$ to a structured latent space $\mathbf{z} \in \mathbb{R}^{128}$ with semantically supervised partitions for tumour volume, location, and shape. Phase 3 encodes a longitudinal Andalusian cohort (42 patients, 137 studies) through the frozen pipeline with optional ComBat harmonisation. Phase 4 fits a three-model Gaussian process hierarchy—Linear Mixed-Effects (LME), Hierarchical GP (H-GP), and Partition-Aware Multi-Output GP (PA-MOGP)—evaluated via Leave-One-Patient-Out Cross-Validation across 33 patients. The pipeline replaces a previously considered Neural ODE approach, which was abandoned due to catastrophic overparameterisation (3,100+ parameters for 112 observation pairs). We report domain gap analysis, LoRA ablation results including dual-domain adaptation experiments, and discuss methodological revisions motivated by empirical findings including the failure of shape disentanglement and the temporal stationarity of meningioma location.

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1 Introduction

1.1 Clinical Motivation

Meningiomas are the most common primary intracranial tumours, accounting for approximately 39% of all central nervous system neoplasms, with an age-adjusted incidence rate of 9.51 per 100,000 population [1]. The majority are classified as WHO Grade I (benign), and a substantial fraction are discovered incidentally on neuroimaging performed for unrelated indications [2]. For these incidentally discovered meningiomas, clinical management hinges on a deceptively simple question: *will this tumour grow?*

Current clinical practice relies on serial MRI surveillance with manual volumetric assessment at intervals of 3–12 months [3, 4]. This approach has three fundamental limitations. First, manual volumetry is labour-intensive and subject to considerable inter-observer variability, with reported coefficient of variation exceeding 15% for small tumours [5]. Second, the decision to intervene (surgery or radiotherapy) is typically made reactively—after significant growth has already occurred—rather than proactively based on predicted trajectories. Third, meningioma growth dynamics are heterogeneous: some tumours remain stable for decades, others exhibit steady linear growth, and a minority display accelerating or saltatory growth patterns [3, 6]. No established imaging biomarker reliably predicts which pattern a given tumour will follow.

Quantitative growth prediction from longitudinal MRI could transform meningioma management by enabling personalised surveillance schedules, early identification of aggressive growth patterns, and evidence-based timing of intervention. However, building such predictive models faces a data bottleneck: longitudinal meningioma MRI datasets are rare, typically small (tens of patients), and collected retrospectively across heterogeneous scanners and protocols.

1.2 Technical Challenge

The central technical challenge is to build a growth prediction model from a longitudinal cohort of only 33 patients with 2–6 MRI studies each (100 total observations), while leveraging cross-sectional data from 1,000+ subjects to learn rich tumour representations. This requires solving three interrelated problems:

1. **Domain adaptation.** Existing brain MRI foundation models are pretrained on glioma data, which differs fundamentally from meningioma in tumour morphology, location, and tissue composition. Direct application of a glioma-trained encoder to meningioma data produces suboptimal feature representations (see Section 4).

2. **Structured representation learning.** Growth prediction requires features that are not only discriminative but also *semantically structured*: tumour volume, spatial loca-

tion, and morphological shape should be disentangled into separate latent subspaces amenable to interpretable temporal modelling.

3. **Small-sample temporal modelling.** With 33 patients and sparse, irregularly-sampled time series, conventional deep learning approaches (recurrent networks, Neural ODEs) are catastrophically overparameterised. The modelling framework must provide calibrated uncertainty and degrade gracefully with limited data.

1.3 Proposed Approach

We address these challenges through a four-phase pipeline that progressively transforms raw multi-modal 3D MRI into patient-specific growth predictions with calibrated uncertainty:

Phase 1 — LoRA Adaptation (Section 5). We adapt BrainSegFounder [7], a SwinUNETR [8] foundation model pretrained via self-supervised learning on 41,400+ brain MRI subjects, to the meningioma domain using Low-Rank Adaptation (LoRA) [9]. LoRA injects trainable low-rank matrices into the encoder's Stages 3–4, adapting high-level features while preserving low-level anatomical representations. Segmentation on BraTS-MEN 2024 [10] serves as the proxy task, with optional auxiliary semantic heads providing multi-task learning signal.

Phase 2 — Supervised Disentangled Projection (Section 6). A lightweight projection network maps frozen encoder features $\mathbf{h} \in \mathbb{R}^{768}$ to a structured latent space $\mathbf{z} \in \mathbb{R}^{128}$ partitioned into semantically supervised subspaces for volume, location, shape, and a residual component. Training combines semantic regression, VICReg-style [11] variance–covariance regularisation, and distance correlation [12] penalties via a four-phase curriculum.

Phase 3 — Cohort Encoding (Section 7). All longitudinal MRI studies from the Andalusian cohort are encoded through the frozen pipeline, producing per-patient latent trajectories. ComBat harmonisation [13, 14] is assessed and applied if significant scanner-induced distributional shift is detected.

Phase 4 — Growth Prediction (Section 8). A three-model Gaussian process hierarchy of increasing complexity—Linear Mixed-Effects (LME), Hierarchical GP (H-GP), and Partition-Aware Multi-Output GP (PA-MOGP)—is fitted on the latent trajectories and evaluated via Leave-One-Patient-Out Cross-Validation (LOPO-CV) across 33 folds. Volume predictions are decoded through the frozen SDP semantic head and compared against observed tumour volumes.

1.4 Contributions

This thesis makes the following contributions:

1. **Domain adaptation analysis.** We provide a quantitative characterisation of the glioma–meningioma domain gap in the BrainSegFounder feature space using Maximum Mean Discrepancy, Centred Kernel Alignment, and semantic linear probes (Section 4).

2. **LoRA adaptation with semantic supervision.** We demonstrate LoRA adaptation of a 3D medical imaging foundation model for meningioma segmentation, with ablations across rank, adapter type (LoRA vs. DoRA), auxiliary semantic heads, and single- vs. dual-domain training (Section 5).

3. **Supervised Disentangled Projection.** We design and theoretically justify a non-generative approach to structured latent space construction, motivated by the impossibility of unsupervised disentanglement [15] and the failure of VAE-based alternatives in our experimental setting (Section 6).

4. **GP hierarchy for latent growth prediction.** We propose a principled replacement for Neural ODE-based temporal modelling, with closed-form inference, calibrated uncertainty, and a parameter count appropriate for the data regime (6–95 parameters vs. 3,100+ for the Neural ODE) (Section 8).

5. **Empirical methodology revision.** We document how empirical findings—including shape disentanglement failure ($R^2 \leq 0.11$), temporal stationarity of meningioma location, and marginal LoRA benefit ($\Delta R^2 = +0.016$)—motivated principled revisions to the pipeline architecture (Section 9).

1.5 Document Structure

The remainder of this report is organised as follows. Section 2 reviews the theoretical foundations underlying each pipeline component. Section 3 describes the datasets, preprocessing, and data splits. Section 4 presents the domain gap analysis motivating LoRA adaptation. Sections 5 to 8 detail Phases 1–4 respectively. Section 9 consolidates quality targets, ablation results, and model comparison. Section 10 discusses limitations and future directions.

2 Background and Related Work

2.1 Foundation Models for Brain MRI

The paradigm of pretraining large encoder–decoder architectures on massive, diverse datasets and subsequently fine-tuning on downstream tasks has transformed medical im-

age analysis [16, 17]. BrainSegFounder [7] instantiates this paradigm for 3D brain MRI segmentation through a three-stage training pipeline:

1. **Self-supervised pretraining.** A SwinUNETR encoder [8, 18] is pretrained on 41,400+ brain MRI subjects from the UK Biobank using a combination of masked image modelling, contrastive learning, and rotation prediction. This stage uses a 96^3 voxel region-of-interest (ROI) and produces general-purpose anatomical features.

2. **Supervised pretraining.** The encoder is paired with the SwinUNETR decoder and trained on the BraTS 2021 glioma segmentation benchmark [19] at 128^3 ROI, producing tumour-specific features.

3. **Per-fold fine-tuning.** Final fine-tuning on BraTS 2021 with cross-validation yields the released checkpoints.

2.1.1 SwinUNETR Architecture

SwinUNETR combines the Swin Transformer [20] with a UNet-style encoder–decoder structure. The encoder consists of a patch embedding layer followed by four transformer stages with shifted window self-attention. For BSF-Tiny (the variant used in this work), the architectural parameters are:

Table 1: SwinUNETR encoder (BSF-Tiny) architecture. Input: $[B, 4, 128^3]$.

Stage	Output shape	Channels	Heads	Blocks
Patch embed	$[B, 48, 64^3]$	48	—	—
Stage 1	$[B, 96, 32^3]$	96	3	2
Stage 2	$[B, 192, 16^3]$	192	6	2
Stage 3	$[B, 384, 8^3]$	384	12	2
Stage 4	$[B, 768, 4^3]$	768	24	2
encoder10	$[B, 768, 4^3]$	768	—	—

The bottleneck layer `encoder10` applies a residual convolutional block preserving the 768-channel, 4^3-spatial resolution from Stage 4. Global Average Pooling (GAP) over the spatial dimensions produces the feature vector $\mathbf{h} \in \mathbb{R}^{768}$ used throughout our pipeline.

The segmentation decoder mirrors the encoder with skip connections, producing a 3-channel sigmoid output following the BraTS convention: Channel 0 encodes Tumour Core (TC), Channel 1 encodes Whole Tumour (WT), and Channel 2 encodes Enhancing Tumour (ET), using hierarchical overlapping labels rather than mutually exclusive classes.

2.2 Low-Rank Adaptation

Low-Rank Adaptation (LoRA) [9] provides parameter-efficient fine-tuning by injecting trainable rank-r decompositions into frozen pretrained weight matrices. For a pretrained

weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d\times k}$, LoRA parameterises the adapted weight as:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A}, \quad \mathbf{B} \in \mathbb{R}^{d\times r},\ \mathbf{A} \in \mathbb{R}^{r\times k}, \tag{1}$$

where $r \ll \min(d, k)$ is the rank, α is a scaling hyperparameter, $\mathbf{A}$ is initialised from $\mathcal{N}(0, \sigma^2)$, and $\mathbf{B} = \mathbf{0}$ so that the adapted model initially reproduces the pretrained model exactly. The number of trainable parameters per adapted layer is $r(d + k)$, compared to dk for full fine-tuning.

The theoretical foundation rests on the observation by Aghajanyan et al. [21] that pretrained language models have a low intrinsic dimensionality: the parameter space can be projected to a low-dimensional subspace without significant loss in fine-tuning performance. LoRA operationalises this insight by constraining the weight update to lie in a rank-r subspace.

DoRA [22] extends LoRA by decomposing the weight update into magnitude and direction components:

$$\mathbf{W} = m \cdot \frac{\mathbf{W}_0 + \mathbf{B}\mathbf{A}}{\|\mathbf{W}_0 + \mathbf{B}\mathbf{A}\|_c}, \tag{2}$$

where m is a learnable magnitude vector and $\|\cdot\|_c$ denotes column-wise normalisation. This decomposition was shown to more closely approximate full fine-tuning behaviour in language models; we evaluate it as an ablation condition (A2, Section 9.2).

2.3 Disentangled Representation Learning

Disentangled representations encode independent generative factors of variation in separate latent dimensions or subspaces. Higgins et al. [23] formalised this via group theory: a representation is disentangled with respect to a group decomposition $G = G_1 \times \cdots \times G_n$ if there exists a corresponding decomposition $Z = Z_1 \oplus \cdots \oplus Z_n$ where each Z_i is affected only by G_i.

The impossibility theorem of Locatello et al. [15] established that unsupervised disentanglement is theoretically impossible without inductive biases: for any generative model producing disentangled representations, there exists another model with identical marginal distribution but entangled latents. Their follow-up work [24] demonstrated that as few as 100 labelled examples suffice to resolve this impossibility through supervision.

These results motivate our Supervised Disentangled Projection (SDP) approach: rather than relying on VAE-based unsupervised disentanglement (which we empirically found to suffer from posterior collapse and KL distortion), we use explicit semantic supervision from segmentation-derived features to anchor each latent partition to a known generative factor.

2.3.1 VICReg Regularisation

VICReg [11] defines three regularisation terms for representation learning. The *variance* term prevents dimensional collapse via a hinge loss on per-dimension standard deviation:

$$\mathcal{L}_{\text{var}}(\mathbf{Z}) = \frac{1}{d}\sum_{j=1}^{d} \max\Big(0,\ \gamma - \sqrt{\text{Var}(z^{(j)}) + \epsilon}\Big), \tag{3}$$

where $\gamma = 1$ is the target standard deviation. The *covariance* term decorrelates dimensions by penalising off-diagonal elements of the batch covariance matrix $\mathbf{C}$:

$$\mathcal{L}_{\text{cov}}(\mathbf{Z}) = \frac{1}{d}\sum_{i \neq j} C_{ij}^2. \tag{4}$$

2.3.2 Distance Correlation

VICReg's covariance penalty captures only linear dependencies. Distance correlation [12] detects all forms of statistical dependence between random vectors X and Y of potentially different dimensions:

$$\text{dCor}(X, Y) = \frac{\text{dCov}(X, Y)}{\sqrt{\text{dVar}(X) \cdot \text{dVar}(Y)}}, \tag{5}$$

where dCov is computed from doubly-centred pairwise distance matrices. Critically, $\text{dCor}(X, Y) = 0$ if and only if $X \perp\!\!\!\perp Y$ (for random vectors with finite first moments), providing a necessary and sufficient condition for statistical independence.

2.3.3 Spectral Normalisation

Spectral normalisation [25] constrains each linear layer $\mathbf{W}$ by dividing by its largest singular value $\sigma(\mathbf{W})$, yielding a weight matrix with spectral norm exactly 1. For an L-layer MLP with 1-Lipschitz activations (e.g., ReLU, GELU), this bounds the global Lipschitz constant: $\text{Lip}(f) \leq \prod_{\ell=1}^{L} 1 = 1$. The contraction property $\|\mathbf{z}_1 - \mathbf{z}_2\| \leq \|\mathbf{h}_1 - \mathbf{h}_2\|$ ensures that encoder feature noise is not amplified through the projection, which is essential for stable downstream GP prediction [26].

2.4 Gaussian Process Regression

A Gaussian process (GP) is a distribution over functions such that any finite collection of function values is jointly Gaussian [27]:

$$f(\mathbf{t}) \sim \mathcal{GP}\Big(m(\mathbf{t}),\ k(\mathbf{t}, \mathbf{t}')\Big), \tag{6}$$

where $m(\mathbf{t})$ is the mean function and $k(\mathbf{t}, \mathbf{t}')$ is the covariance (kernel) function. Given n observations $\mathbf{y} = f(\mathbf{t}) + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$, the predictive distribution at test inputs

$\mathbf{t}_*$ is:

$$\mathbf{f}_*|\mathbf{t}_*, \mathbf{t}, \mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*), \tag{7}$$

$$\boldsymbol{\mu}_* = m(\mathbf{t}_*) + \mathbf{K}_*^\top(\mathbf{K} + \sigma_n^2\mathbf{I})^{-1}(\mathbf{y} - m(\mathbf{t})), \tag{8}$$

$$\boldsymbol{\Sigma}_* = \mathbf{K}_{**} - \mathbf{K}_*^\top(\mathbf{K} + \sigma_n^2\mathbf{I})^{-1}\mathbf{K}_*, \tag{9}$$

where $\mathbf{K} = k(\mathbf{t}, \mathbf{t})$, $\mathbf{K}_* = k(\mathbf{t}, \mathbf{t}_*)$, and $\mathbf{K}_{**} = k(\mathbf{t}_*, \mathbf{t}_*)$.

2.4.1 Matérn Kernels

The Matérn family of kernels provides control over the smoothness of the GP sample paths via the parameter ν:

$$k_\nu(\Delta t) = \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}\,|\Delta t|}{\ell} \right)^\nu K_\nu\left(\frac{\sqrt{2\nu}\,|\Delta t|}{\ell} \right), \tag{10}$$

where ℓ is the length-scale, σ_f^2 the signal variance, and K_ν the modified Bessel function. For $\nu = 5/2$, the kernel yields twice-differentiable sample paths:

$$k_{5/2}(\Delta t) = \sigma_f^2 \left(1 + \frac{\sqrt{5}\,|\Delta t|}{\ell} + \frac{5\,\Delta t^2}{3\ell^2} \right) \exp\left(-\frac{\sqrt{5}\,|\Delta t|}{\ell} \right). \tag{11}$$

This is our default temporal kernel for volume dynamics, encoding the prior belief that tumour volume evolves smoothly over time.

2.4.2 Multi-Output GPs and the Intrinsic Coregionalisation Model

Multi-Output Gaussian Processes (MOGPs) extend scalar GPs to vector-valued functions $\mathbf{f} : \mathbb{R} \to \mathbb{R}^D$, capturing correlations across output dimensions. The Intrinsic Coregionalisation Model (ICM) [28] parameterises the cross-output covariance as a Kronecker product:

$$\mathbf{K}(\mathbf{t}, \mathbf{t}') = \mathbf{B} \otimes k_t(\mathbf{t}, \mathbf{t}'), \tag{12}$$

where $\mathbf{B} \in \mathbb{R}^{D \times D}$ is a positive semi-definite coregionalisation matrix encoding inter-output correlations, and k_t is a scalar temporal kernel shared across outputs. The Linear Model of Coregionalisation (LMC) generalises this to a sum over Q latent functions:

$$\mathbf{K}(\mathbf{t}, \mathbf{t}') = \sum_{q=1}^{Q} \mathbf{B}_q \otimes k_q(\mathbf{t}, \mathbf{t}'). \tag{13}$$

2.5 Related Work

2.5.1 Brain Tumour Growth Prediction

Tumour growth prediction from longitudinal MRI has been addressed through several paradigms. Classical approaches fit parametric growth models (exponential, Gompertz, logistic) to volume trajectories [29, 3]. Reaction–diffusion models [30, 31] incorporate spatial diffusion but require tissue-specific parameterisation and are computationally expensive.

Recent deep learning approaches include spatio-temporal convolutional LSTMs [32] achieving 83.2% Dice on glioma growth prediction, GP-GAN [33] combining Gaussian processes with GANs, and neural fields conditioned on time [34] for vestibular schwannoma growth. Puglisi et al. [35] demonstrated that a linear mixed-effects model in a learned latent space produces clinically meaningful progression predictions for brain atrophy, trained on 11,730 MRIs.

For meningiomas specifically, most existing work is limited to retrospective volumetric analysis [3, 4, 6]. To our knowledge, no prior work has applied foundation model-derived latent spaces with GP-based temporal modelling to meningioma growth prediction.

2.5.2 Latent Space Temporal Prediction

The Joint Embedding Predictive Architecture (JEPA) framework [36] advocates predicting in representation space rather than pixel space, avoiding the waste of capacity on irrelevant details. I-JEPA [37] and V-JEPA [38] validated this for images and video respectively. Longitudinal Self-Supervised Learning (LSSL) [39] disentangles brain-age factors from latent MRI representations on 811 ADNI subjects. Our pipeline aligns with this "predict in latent space" principle while using GPs instead of neural predictors, providing closed-form uncertainty quantification for clinical decision support.

2.5.3 Domain Adaptation in Medical Imaging

Domain adaptation for medical image analysis has been addressed through adversarial training [40], style transfer [41], and more recently through parameter-efficient fine-tuning of foundation models [42]. Ben-David et al. [43] established the theoretical framework relating domain divergence to target risk. Our work contributes empirical evidence on LoRA adaptation for 3D medical imaging foundation models, including dual-domain training that attempts to maintain glioma competence while acquiring meningioma specialisation.

3 Data

This section describes the three datasets, the unified HDF5 storage schema, data splits, and the preprocessing pipeline.

3.1 Datasets

3.1.1 BraTS-MEN 2024 (Cross-Sectional Meningioma)

The BraTS Meningioma 2024 challenge dataset [10] comprises 1,000 subjects with pre-operative multi-parametric MRI (T1, T1ce, T2, FLAIR) and expert manual segmentations. Each subject has a single study (timepoint). Segmentation labels follow the BraTS convention: 0 (background), 1 (enhancing tumour, ET), 2 (non-enhancing tumour, NET), and 3 (cystic component). This dataset serves as the primary training source for Phases 1 and 2.

3.1.2 BraTS-GLI 2021 (Longitudinal Glioma)

The BraTS Glioma 2021 challenge dataset [19] contains 1,350 scans from 613 patients (1–10 timepoints per patient) with the same four MRI modalities. Glioma labels use: 0 (background), 1 (necrotic tumour core, NETC), 2 (peritumoral oedema, SNFH), 3 (enhancing tumour, ET), and 4 (resection cavity, RC). This dataset is used for domain gap analysis (Section 4) and dual-domain LoRA training (Section 5.6).

3.1.3 Andalusian Longitudinal Cohort (MenGrowth)

A retrospective longitudinal cohort of 42 meningioma patients (137 total studies) collected across hospitals in Andalusia, Spain. After quality control, 33 patients (100 studies) with 2–6 timepoints each are retained for growth prediction. Observation statistics are summarised in Table 2.

Table 2: Andalusian longitudinal cohort statistics (after QC).

Statistic	**Value**
Total patients (N)	33
Total studies (S)	100
Mean studies/patient ($\bar{n}$)	3.03
Std studies/patient (σ_n)	1.17
Range	[2, 6]
Patients with exactly 2 studies	19 (57.6%)

3.2 Unified HDF5 Schema

All three datasets are stored in a unified HDF5 v2.0 schema, enabling a single reader code path (`BraTSDatasetH5`) regardless of dataset type. The schema structure is:

Table 3: Unified HDF5 v2.0 schema. N_s: number of scans, N_p: number of patients.

Key	Shape/Type	Description
`attrs`	dict	{`n_scans`, `n_patients`, `roi_size`, `spacing`, `channel_order`, `version`="2.0", `dataset_type`, `domain`}
`images`	$[N_s, 4, 192^3]$ float32	Multi-modal MRI volumes
`segs`	$[N_s, 1, 192^3]$ int8	Segmentation masks
`scan_ids`	$[N_s]$ str	Unique scan identifiers
`patient_ids`	$[N_s]$ str	Patient identifiers
`timepoint_idx`	$[N_s]$ int32	Per-patient timepoint index
`semantic/`	group	Precomputed features: `volume` $[N_s, 4]$, `location` $[N_s, 3]$, `shape` $[N_s, 3]$
`longitudinal/`	group	CSR structure: `patient_offsets` $[N_p + 1]$, `patient_list` $[N_p]$
`splits/`	group	Patient-level index arrays: `lora_train`, `lora_val`, `test`
`metadata/`	group	`grade`, `age`, `sex` (when available)

For cross-sectional data (BraTS-MEN), the longitudinal structure is trivial: each patient has exactly one scan at timepoint 0, so $N_s = N_p$ and `patient_offsets` $= [0, 1, 2, \ldots, N]$. This design ensures that all timepoints of a patient are assigned to the same split, preventing patient-level data leakage.

3.3 Data Splits

Data splitting is performed at the patient level with seed 42 for reproducibility. Table 4 summarises the splits for each dataset.

Table 4: Data splits. BraTS-MEN splits are canonical; server configurations may differ slightly (see Section 5.7).

Split	BraTS-MEN	BraTS-GLI (patients)	MenGrowth
`lora_train`	750	430 (∼950 scans)	23
`lora_val`	100	60 (∼130 scans)	3
`test`	150	123 (∼270 scans)	7
Total	1,000	613 (1,350 scans)	33 (100 scans)

For Phase 2 (SDP), the canonical specification defines an additional `sdp_train` split

of 225 subjects; however, the actual implementation uses the `lora_train` pool since the SDP operates on frozen encoder features and there is no gradient contamination concern.

3.4 Preprocessing Pipeline

All volumes are spatially preprocessed to 192^3 voxels at $1\,\mathrm{mm}^3$ isotropic spacing during HDF5 conversion. Intensity normalisation is performed at runtime. The critical channel ordering convention is $[\mathrm{FLAIR}, \mathrm{T1ce}, \mathrm{T1}, \mathrm{T2}]$, matching BrainSegFounder's training protocol—incorrect ordering causes near-zero Dice (~ 0.00) even on the glioma training domain.

3.4.1 Training Transforms

The training pipeline follows the BrainSegFounder protocol with minor adaptations:

1. `CropForegroundd`: Crop to the non-zero brain region with `k_divisible`$= [128, 128, 128]$.
2. `SpatialPadd`: Pad to at least 128^3.
3. `RandSpatialCropd`: Random crop to 128^3.
4. `NormalizeIntensityd`: Per-channel, nonzero-voxel normalisation.
5. Data augmentation: random flips (3 axes, $p = 0.5$ each), intensity scaling (factor 0.1, $p = 1.0$), intensity shift (offset 0.1, $p = 1.0$).

3.4.2 Feature Extraction Transforms

For feature extraction and evaluation, a larger 192^3 ROI is used (via `ResizeWithPadOrCropd`) to ensure 100% tumour containment—the 128^3 centre crop captures only 38.8% of meningioma tumours fully due to their dural-based, peripheral location.

3.4.3 Inference Transforms

Full-resolution inference uses MONAI's sliding window inference with 128^3 patches, 50% overlap, and Gaussian blending.

3.5 Semantic Features

Semantic features are precomputed from segmentation masks and stored in the HDF5 files. Three categories of features are extracted:

Volume (4 dims): $[\log(V_{\mathrm{total}} + 1), \log(V_{\mathrm{NCR}} + 1), \log(V_{\mathrm{ED}} + 1), \log(V_{\mathrm{ET}} + 1)]$, where volumes are in mm^3. The log-transform stabilises the heavy-tailed distribution.

Location (3 dims): Centroid coordinates (c_z, c_y, c_x) in mm, computed as the centre of mass of the tumour segmentation mask.

Shape (3 dims): Sphericity $\in (0, 1]$, log(surface area), and solidity (ratio of tumour volume to convex hull volume), computed via `compute_shape_array()`.

Remark 1. *Following the R1 methodology revision (Section 9.4), the volume target is reduced to a single whole-tumour log-volume* $v^* = \log(V_{WT} + 1) \in \mathbb{R}^1$*, and shape features are dropped from the SDP due to persistent low* R^2 *(*≤ 0.11 *across all conditions).*

3.6 Segmentation Label Conversion

Both datasets are converted to a unified 3-channel hierarchical overlapping representation for the segmentation loss:

$$\mathrm{TC} = \{x : \mathrm{label}(x) \in S_{\mathrm{TC}}\}, \tag{14}$$

$$\mathrm{WT} = \{x : \mathrm{label}(x) > 0\}, \tag{15}$$

$$\mathrm{ET} = \{x : \mathrm{label}(x) \in S_{\mathrm{ET}}\}, \tag{16}$$

where S_{TC} and S_{ET} are domain-dependent. For MEN: $S_{\mathrm{TC}} = \{1, 2\}$, $S_{\mathrm{ET}} = \{1\}$. For GLI: $S_{\mathrm{TC}} = \{1, 3, 4\}$, $S_{\mathrm{ET}} = \{3\}$. This conversion is implemented in `segmentation.py._convert_target()` and ensures that the segmentation decoder output semantics are consistent across domains.

4 Domain Gap Analysis

BrainSegFounder was pretrained and fine-tuned on glioma data. Before adapting it to the meningioma domain, we quantify the distributional shift in the frozen encoder's feature space to establish the empirical necessity of domain adaptation.

4.1 Feature Extraction Protocol

Features are extracted from the frozen BrainSegFounder encoder using the `FeatureExtractor` pipeline:

$$\mathbf{h} = \mathrm{GAP}\Big(\texttt{encoder10}\Big(\mathrm{SwinViT}(\mathbf{x})\Big)\Big) \in \mathbb{R}^{768}, \tag{17}$$

where $\mathbf{x} \in \mathbb{R}^{4\times 192^3}$ is the input volume and GAP denotes Global Average Pooling over the 4^3 (or 6^3 for 192^3 input) spatial dimensions. We extract features from 200 random BraTS-GLI subjects and 200 BraTS-MEN subjects from the test and SDP-train splits.

4.2 Domain Gap Metrics

4.2.1 Maximum Mean Discrepancy

The Maximum Mean Discrepancy (MMD) [44] measures the distance between two distributions in a reproducing kernel Hilbert space. Given samples $\{\mathbf{h}_i^{\text{GLI}}\}_{i=1}^n$ and $\{\mathbf{h}_j^{\text{MEN}}\}_{j=1}^m$, the unbiased estimator is:

$$\widehat{\text{MMD}}^2 = \frac{1}{n(n-1)} \sum_{i \neq i'} k(\mathbf{h}_i, \mathbf{h}_{i'}) + \frac{1}{m(m-1)} \sum_{j \neq j'} k(\mathbf{h}_j, \mathbf{h}_{j'}) - \frac{2}{nm} \sum_{i,j} k(\mathbf{h}_i, \mathbf{h}_j), \tag{18}$$

where k is a Gaussian kernel with bandwidth σ set to the median pairwise distance (the median heuristic). Statistical significance is assessed via a permutation test with $n_{\text{perm}} = 1{,}000$ permutations.

4.2.2 Centred Kernel Alignment

CKA [45] measures representational similarity between two feature matrices $\mathbf{H}_1 \in \mathbb{R}^{n \times d_1}$ and $\mathbf{H}_2 \in \mathbb{R}^{n \times d_2}$ via:

$$\text{CKA}(\mathbf{H}_1, \mathbf{H}_2) = \frac{\|\bar{\mathbf{H}}_1^\top \bar{\mathbf{H}}_2\|_F^2}{\|\bar{\mathbf{H}}_1^\top \bar{\mathbf{H}}_1\|_F \cdot \|\bar{\mathbf{H}}_2^\top \bar{\mathbf{H}}_2\|_F}, \tag{19}$$

where $\bar{\mathbf{H}}$ denotes column-centred matrices. CKA $\in [0, 1]$, with 1 indicating identical representations (up to linear transformation).

4.2.3 Linear Probes

Following Alain and Bengio [46], we train Ridge regression probes from frozen features $\mathbf{h}$ to semantic targets (volume, location, shape). Cross-validated R^2 quantifies the linear decodability of each semantic factor. Low R^2 on the target domain indicates that the frozen encoder has not learned meningioma-relevant features.

4.3 Results

Table 5: Domain gap metrics between frozen BrainSegFounder features on BraTS-GLI vs. BraTS-MEN. Results confirm significant distributional shift motivating domain adaptation.

Metric	**Value**	**Interpretation**
MMD^2 (GLI vs. MEN)	*[to be filled]*	> 0 confirms shift
MMD p-value (permutation test)	*[to be filled]*	< 0.05 confirms significance
CKA (GLI vs. MEN)	*[to be filled]*	Low $\Rightarrow$ different structure
Linear probe R^2 (volume, MEN)	*[to be filled]*	Baseline decodability
Linear probe R^2 (location, MEN)	*[to be filled]*	Baseline decodability
Linear probe R^2 (shape, MEN)	*[to be filled]*	Baseline decodability

Figure placeholder: UMAP of frozen BrainSegFounder features. GLI (orange) vs. MEN (blue). Expected: clear separation.

Figure 1: UMAP visualisation of frozen BrainSegFounder encoder features ($\mathbf{h} \in \mathbb{R}^{768}$) for BraTS-GLI (orange, $n = 200$) and BraTS-MEN (blue, $n = 200$). The visible separation confirms the domain gap motivating LoRA adaptation.

The domain gap analysis confirms that the frozen glioma-trained encoder produces features with significant distributional shift when applied to meningioma data. The MMD permutation test rejects the null hypothesis of identical distributions, and the linear probe R^2 values on MEN data are substantially lower than within-domain GLI probes, indicating that the frozen features inadequately capture meningioma-specific semantic content. These findings motivate the LoRA adaptation described in Section 5.

4.4 Effective Rank Analysis

The effective rank of the feature matrix, defined as the exponential of the spectral entropy:

$$\mathrm{erank}(\mathbf{H}) = \exp\left(-\sum_{i=1}^{d} \bar{s}_i \log \bar{s}_i\right), \quad \bar{s}_i = \frac{s_i}{\sum_j s_j}, \tag{20}$$

where s_i are the singular values of $\mathbf{H}$, measures the distributional uniformity of the representation. The frozen BrainSegFounder encoder achieves an effective rank of approximately 47.6 on MEN data, indicating moderate utilisation of the 768-dimensional feature space.

5 Phase 1: LoRA Encoder Adaptation

5.1 Adaptation Strategy

We adapt the frozen BrainSegFounder encoder to the meningioma domain via LoRA (Section 2.2), injecting low-rank adapter matrices into the self-attention projections of Stages 3–4. Stages 0–2 remain frozen to preserve low-level anatomical features learned from the UK Biobank SSL pretraining. Segmentation on BraTS-MEN 2024 serves as the proxy task.

5.1.1 Target Modules

LoRA adapters are injected into the **Q**, **K**, **V** projection matrices within the multi-head self-attention blocks of Stages 3 and 4:

$$\texttt{swinViT.layers3.0.blocks.\{0,1\}.attn.qkv.weight}, \tag{21}$$

$$\texttt{swinViT.layers4.0.blocks.\{0,1\}.attn.qkv.weight}. \tag{22}$$

This yields 4 adapted layers, each with rank-r decompositions for Q, K, and V (combined as a single QKV projection in the Swin implementation). The total number of trainable LoRA parameters is approximately 197K for $r = 8$, compared to 62M total model parameters ($< 0.32\%$).

5.1.2 LoRA Hyperparameters

Table 6: LoRA configuration (selected via rank ablation, see Section 5.5).

Parameter	Value	Rationale
Rank (r)	8	Optimal from ablation over $r \in \{2, 4, 8, 16, 32\}$
Scale (α)	16	Effective scale $\alpha/r = 2.0$
Dropout	0.1	Regularisation
Target stages	3, 4	High-level features; stages 0–2 frozen
Target modules	QKV	All attention projections

5.2 Auxiliary Semantic Heads

To enrich the encoder's feature space for downstream SDP training, we attach auxiliary semantic regression heads that predict tumour properties directly from GAP-pooled encoder features:

$$\hat{\mathbf{y}}_p = \pi_p\Big(\mathrm{GAP}(\texttt{encoder10}(\mathbf{x}))\Big), \quad p \in \{\mathrm{vol}\}, \tag{23}$$

where $\pi_{\mathrm{vol}} : \mathbb{R}^{768} \to \mathbb{R}^1$ is a linear head predicting log whole-tumour volume. The auxiliary loss is:

$$\mathcal{L}_{\mathrm{aux}} = \lambda_{\mathrm{aux}} \cdot \|\hat{\mathbf{y}}_{\mathrm{vol}} - \mathbf{y}_{\mathrm{vol}}\|_2^2. \tag{24}$$

Following the R1 revision (Section 9.4), auxiliary heads for location and shape have been removed; only the volume head is retained. The auxiliary loss is activated after epoch 5 with a linear warmup over 10 epochs ($\lambda_{\mathrm{aux}} = 0.1$ at full strength).

Remark 2. *The auxiliary heads are discarded after Phase 1. Their sole purpose is to provide multi-task learning signal during encoder adaptation, encouraging the encoder to produce features from which tumour volume can be linearly decoded.*

5.3 Training Protocol

Table 7: Phase 1 training configuration.

Parameter	Value	Rationale
Epochs	150	With early stopping (patience 25)
LR (LoRA)	10^{-4}	Standard for LoRA adaptation
LR (decoder)	5×10^{-4}	Larger for pretrained decoder
LR (aux heads)	10^{-3}	Small heads, fast convergence
Optimizer	AdamW	Decoupled weight decay
Weight decay	10^{-5}	Light regularisation
Scheduler	Cosine decay, 10-epoch warmup	Smooth convergence
Batch size	2 (per GPU)	Memory-constrained by 128^3 input
Gradient accumulation	4 steps	Effective batch size 8
Gradient clipping	$\|\mathbf{g}\|_2 \leq 1.0$	Stability
Precision	bf16-mixed	Memory efficiency on A100

The segmentation loss combines Dice loss and cross-entropy with equal weight:

$$\mathcal{L}_{\text{seg}} = \lambda_{\text{dice}}\, \mathcal{L}_{\text{Dice}} + \lambda_{\text{CE}}\, \mathcal{L}_{\text{CE}}, \quad \lambda_{\text{dice}} = \lambda_{\text{CE}} = 1.0. \tag{25}$$

The total training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + w(e)\, \mathcal{L}_{\text{aux}}, \tag{26}$$

where $w(e) = \lambda_{\text{aux}} \cdot \min(1, \max(0, (e-5)/10))$ is the warmup schedule for the auxiliary loss starting at epoch 5.

5.4 Post-Training: LoRA Merge and Checkpoint Export

After training, LoRA weights are absorbed into the base model weights:

$$\mathbf{W}_{\text{merged}} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A}. \tag{27}$$

The segmentation decoder and auxiliary heads are then discarded, yielding a merged encoder checkpoint (`phase1_encoder_merged.pt`) containing only the SwinViT and encoder1–10 weights. All encoder parameters are frozen for subsequent phases.

5.5 Ablation Results

5.5.1 Rank Sweep (A1)

We evaluate LoRA ranks $r \in \{2, 4, 8, 16, 32\}$ on segmentation Dice and linear probe R^2 for volume prediction.

Table 8: LoRA rank ablation (A1). Dice scores on `lora_val` and linear probe R^2 for volume on test features.

Rank	**LoRA params**	**Dice (WT)**	**Dice (TC)**	**Dice (ET)**	**Vol R^2**
Frozen	0	*[baseline]*	*[baseline]*	*[baseline]*	*[baseline]*
2	∼49K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
4	∼99K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
8	∼197K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
16	∼394K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
32	∼787K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*

Figure placeholder: LoRA rank ablation curves.
Left: Dice vs. rank. Right: Probe R^2 vs. rank.

Figure 2: LoRA rank ablation (A1). Left panel shows validation Dice scores per sub-region as a function of LoRA rank. Right panel shows linear probe R^2 for volume prediction. Rank $r = 8$ achieves the best trade-off between capacity and parameter efficiency.

5.5.2 LoRA vs. DoRA (A2)

DoRA (Equation (2)) is evaluated as an alternative adapter type at $r = 8$.

Table 9: LoRA vs. DoRA comparison (A2) at rank 8.

Adapter	**Dice (WT)**	**Vol R^2 (linear)**	**Vol R^2 (RBF)**
LoRA ($r = 8$)	*[to fill]*	*[to fill]*	*[to fill]*
DoRA ($r = 8$)	*[to fill]*	*[to fill]*	*[to fill]*

5.6 Dual-Domain Adaptation

A key scientific question is whether a single LoRA-adapted encoder can produce semantically structured features for both glioma and meningioma simultaneously. Dual-domain training combines BraTS-MEN (∼750 studies) and BraTS-GLI (∼950 scans) via a `ConcatDataset` with domain-balanced `WeightedRandomSampler` (50/50 domain weight).

5.6.1 Empirical Findings

Table 10: Single-domain vs. dual-domain LoRA adaptation results.

	MEN Dice	**Vol R^2**	**CV Vol R^2**	**MMD^2**	**CKA**	**erank**
Frozen baseline	0.52	0.56	—	0.118	0.002	47.6
MEN LoRA r=8	0.48	0.57	—	0.116	0.002	45.8
Dual LoRA r=8	0.57	0.62	0.712	0.037	0.017	25.4

The dual-domain encoder achieves the best volume probe R^2 and reduces the domain gap (MMD^2: 0.118 → 0.037, a 68.6% reduction), but at the cost of a substantial drop in effective rank (47.6 → 25.4, a 46.6% reduction), indicating representation compression.

5.6.2 Critical Finding: Cross-Domain Transfer Failure

Despite the reduced domain gap metrics, cross-domain GP probes reveal a critical limitation: training probes on GLI features and evaluating on MEN features yields $R^2 \approx -0.01$ for volume, indicating that the shared feature space does not support cross-domain semantic transfer. Furthermore, MEN segmentation Dice degrades from 0.64 to 0.36 under dual training, indicating catastrophic interference rather than beneficial transfer.

Remark 3. *The dual-domain results provide important negative evidence: domain gap reduction (measured by MMD, CKA) does not guarantee cross-domain semantic compatibility. The effective rank drop from 47.6 to 25.2 suggests that the encoder resolves the domain conflict by compressing both domains into a lower-dimensional shared subspace that loses domain-specific semantic content.*

5.6.3 LoRA Marginality

A surprising finding is the marginal benefit of LoRA adaptation for semantic probes: the frozen baseline achieves volume $R^2 = 0.794$, while the best LoRA condition improves this by only $\Delta R^2 = +0.016$ (within noise). This suggests that BrainSegFounder's SSL pretraining already produces features with reasonable meningioma volume decodability, and the primary benefit of LoRA is in segmentation quality rather than downstream feature quality.

5.7 Configuration

All LoRA experiments are configured via YAML files under `experiments/lora/config/`. Server configurations use 250/50/200/500 splits (lora_train/lora_val/probe_train/test), differing slightly from the canonical specification. Four conditions are defined: LoRA/-DoRA × semantic/no-semantic heads.

6 Phase 2: Supervised Disentangled Projection

6.1 Motivation

The encoder features $\mathbf{h} \in \mathbb{R}^{768}$ are high-dimensional and unstructured: semantic factors (volume, location, shape) are entangled across all 768 dimensions. Downstream GP-based growth prediction requires features that are (i) *semantically interpretable*: tumour volume should be decodable from a known subset of latent dimensions; (ii) *disentangled*: volume, location, and shape should occupy independent subspaces; (iii) *low-dimensional*: the GP covariance matrix scales as $O(D^2)$ in the output dimensionality D.

The Supervised Disentangled Projection (SDP) addresses all three requirements through a lightweight MLP trained on frozen encoder features with a composite loss combining semantic supervision, collapse prevention, and independence regularisation.

6.1.1 Why Not a VAE?

Early experiments with a Semi-Supervised VAE (SemiVAE) were abandoned due to three failure modes: (i) posterior collapse, where latent dimensions become uninformative despite extensive hyperparameter tuning (7-phase curriculum across 600 epochs); (ii) KL distortion, where the $D_{\mathrm{KL}}(q\|p)$ penalty actively minimises the mutual information $I(x; z)$ between input and representation, destroying the rich foundation model features; and (iii) reconstruction burden, where the decoder wastes capacity on pixel-level fidelity irrelevant to downstream growth prediction.

The SDP replaces the generative objective with a discriminative one that directly optimises for the properties needed by the GP temporal model, following the theoretical mandate of Locatello et al. [15]: supervised disentanglement resolves the impossibility of unsupervised approaches.

6.2 Architecture

The SDP consists of a projection network g_θ and per-partition semantic heads $\{\pi_p\}$:

$$g_\theta : \mathbb{R}^{768} \to \mathbb{R}^{128}, \quad g_\theta(\mathbf{h}) = \mathrm{SN}(\mathbf{W}_2)\, \sigma\Big(\mathrm{SN}(\mathbf{W}_1)\, \mathrm{LN}(\mathbf{h}) + \mathbf{b}_1\Big) + \mathbf{b}_2, \tag{28}$$

where LN is LayerNorm, σ is GELU with dropout $p = 0.1$, and SN$(\cdot)$ denotes spectral normalisation applied to *both* linear layers (Section 2.3). The architecture is:

$$\underbrace{\mathbf{h} \in \mathbb{R}^{768}}_{\text{frozen encoder}} \xrightarrow{\text{LN}} \xrightarrow{\text{SN-Linear(768,512)}} \xrightarrow{\text{GELU + Drop}} \xrightarrow{\text{SN-Linear(512,128)}} \underbrace{\mathbf{z} \in \mathbb{R}^{128}}_{\text{structured latent}} \tag{29}$$

The total parameter count is approximately 459K ($768 \times 512 + 512 + 512 \times 128 + 128$).

6.3 Latent Partition Structure

The 128-dimensional latent vector is partitioned into semantically supervised and unsupervised subspaces:

Table 11: Original SDP partition structure ($3 + 1$ partitions). See Section 9.4 for the R1-revised $1 + 1$ structure.

Partition	**Dims**	**Indices**	**Target dim**	**Semantic head**
$\mathbf{z}_{\text{vol}}$ (volume)	24	$[0, 24)$	4	$\pi_{\text{vol}} : \mathbb{R}^{24} \to \mathbb{R}^4$
$\mathbf{z}_{\text{loc}}$ (location)	8	$[24, 32)$	3	$\pi_{\text{loc}} : \mathbb{R}^8 \to \mathbb{R}^3$
$\mathbf{z}_{\text{shape}}$ (shape)	12	$[32, 44)$	6	$\pi_{\text{shape}} : \mathbb{R}^{12} \to \mathbb{R}^6$
$\mathbf{z}_{\text{res}}$ (residual)	84	$[44, 128)$	—	None (unsupervised)
Total	128	$[0, 128)$	13	—

Each supervised partition p has a linear semantic head $\pi_p : \mathbb{R}^{d_p} \to \mathbb{R}^{k_p}$ that predicts ground-truth semantic features extracted from segmentation masks. The residual partition captures information not explained by the three supervised factors.

Remark 4 (R1 Revision)**.** *Following empirical results (shape $R^2 \leq 0.11$, temporally static meningioma location), the partition structure is revised to $\mathbf{z} = [\mathbf{z}_{vol} \in \mathbb{R}^{32} \mid \mathbf{z}_{res} \in \mathbb{R}^{96}]$ with a single volume head $\pi_{vol} : \mathbb{R}^{32} \to \mathbb{R}^1$ predicting $\log(V_{WT} + 1)$. Location is relocated to a static GP covariate (Section 9.4).*

6.4 Composite Loss Function

The SDP training objective combines four terms:

$$\mathcal{L}_{\text{SDP}} = \underbrace{\sum_{p \in \mathcal{P}} \lambda_p \, \|\hat{\mathbf{y}}_p - \mathbf{y}_p\|_2^2}_{\text{Semantic regression}} + \underbrace{\lambda_{\text{var}} \, \mathcal{L}_{\text{var}}(\mathbf{z})}_{\text{Collapse prevention}} + \underbrace{\lambda_{\text{cov}} \, \mathcal{L}_{\text{cov}}(\mathbf{z})}_{\text{Linear independence}} + \underbrace{\lambda_{\text{dCor}} \sum_{\substack{i,j \in \mathcal{P} \\ i<j}} \text{dCor}(\mathbf{z}_i, \mathbf{z}_j)}_{\text{Nonlinear independence}}, \tag{30}$$

where $\mathcal{P}$ is the set of supervised partitions, $\hat{\mathbf{y}}_p = \pi_p(\mathbf{z}_p)$, and each loss term serves a distinct purpose.

6.4.1 Semantic Regression

For each supervised partition, MSE loss anchors the partition's information content to a known semantic factor:

$$\mathcal{L}_p^{\text{sem}} = \frac{1}{k_p} \|\pi_p(\mathbf{z}_p) - \mathbf{y}_p\|_2^2. \tag{31}$$

Targets $\mathbf{y}_p$ are normalised to zero mean and unit variance using statistics computed on the training pool only (Section 6.7).

6.4.2 Variance Hinge Loss (Collapse Prevention)

The VICReg variance term (Equation (3)) is applied to all 128 dimensions of $\mathbf{z}$, ensuring no dimension collapses to a constant. The threshold $\gamma = 1.0$ targets unit standard deviation per dimension.

6.4.3 Covariance Penalty (Linear Independence)

Cross-partition covariance is penalised:

$$\mathcal{L}_{\text{cov}} = \frac{1}{d^2} \sum_{\substack{i,j: \\ \text{part}(i) \neq \text{part}(j)}} C_{ij}^2, \tag{32}$$

where $\mathbf{C}$ is the batch covariance matrix and $\text{part}(i)$ denotes the partition containing dimension i. Only cross-partition off-diagonal elements are penalised; within-partition correlations are permitted.

6.4.4 Distance Correlation (Nonlinear Independence)

For each pair of supervised partitions (i, j), the distance correlation $\text{dCor}(\mathbf{z}_i, \mathbf{z}_j)$ (Equation (5)) captures nonlinear dependencies not detected by the covariance penalty. The empirical distance correlation is computed from $O(N^2)$ pairwise distance matrices, which for $N = 800$ (full-batch training) requires approximately 30 MB of GPU memory per partition pair.

6.5 Curriculum Training Schedule

Following the capacity-annealing principle from the disentanglement literature [47], losses are activated in a staged curriculum:

Table 12: SDP curriculum training schedule (4 phases).

Phase	Epochs	Active losses
Warm-up	0–9	$\mathcal{L}_{\text{var}}$ only
Semantic	10–39	$+\mathcal{L}_{\text{vol}}, \mathcal{L}_{\text{loc}}, \mathcal{L}_{\text{shape}}$
Independence	40–59	$+\mathcal{L}_{\text{cov}}, \mathcal{L}_{\text{dCor}}$
Full	60–100	All losses at full strength

The rationale is that semantic regression must first establish the partition structure before independence regularisation is imposed. Premature independence penalties can collapse dimensions before they have acquired semantic content.

6.6 Hyperparameters

Table 13: SDP training hyperparameters.

Parameter	Value	Rationale
λ_{vol}	20.0	Highest weight; volume is primary target
λ_{loc}	12.0	Moderate weight
λ_{shape}	15.0	Higher than location (harder to learn)
λ_{cov}	5.0	Cross-partition linear independence
λ_{var}	5.0	Collapse prevention
λ_{dCor}	2.0	Nonlinear independence
Learning rate	10^{-3}	Standard for small MLPs
Optimizer	AdamW	Decoupled weight decay
Weight decay	0.01	Regularisation
Epochs	100	Sufficient for convergence
Batch size	800 (full-batch)	Exact covariance/dCor estimates
Scheduler	Cosine, 5-epoch warmup	Smooth convergence

The full-batch training is computationally trivial since the SDP operates on precomputed 768-dimensional feature vectors, not 3D volumes. Full-batch provides exact (not estimated) covariance and distance correlation statistics.

6.7 Normalisation Scope

Target normalisation statistics (μ, σ) are computed on the training pool (800 subjects) only and applied without recomputation to validation (100), test (150), and the Andalusian cohort. This prevents information leakage and ensures consistent scaling across all downstream uses of the semantic heads.

6.8 Lipschitz Continuity Analysis

With spectral normalisation on both linear layers and GELU activation (Lipschitz constant ≈ 1.13), the global Lipschitz constant of the SDP is bounded by:

$$\mathrm{Lip}(g_\theta) \leq \underbrace{1}_{\mathrm{LN}} \cdot \underbrace{1}_{\mathrm{SN}(\mathbf{W}_1)} \cdot \underbrace{1.13}_{\mathrm{GELU}} \cdot \underbrace{1}_{\mathrm{SN}(\mathbf{W}_2)} \approx 1.13. \tag{33}$$

This ensures that the projection is a near-contraction: $\|\mathbf{z}_1 - \mathbf{z}_2\| \leq 1.13\,\|\mathbf{h}_1 - \mathbf{h}_2\|$. Encoder feature noise is not amplified through the projection, yielding well-conditioned inputs for downstream GP models.

6.9 Quality Targets

Table 14: SDP quality targets (validation set).

Metric	**Minimum**	**Target**	**Blocking?**
Volume R^2	0.80	0.90	Yes
Location R^2	0.85	0.95	Yes
Shape R^2	0.30	0.40	Yes
Max cross-partition correlation	< 0.30	< 0.20	Yes
$\mathrm{dCor}(\mathrm{vol}, \mathrm{loc})$	< 0.20	< 0.10	Yes
Dims with variance > 0.5	$\geq 85\%$	$\geq 95\%$	Yes

Remark 5. *The shape R^2 target of 0.30 was not achieved in any experimental condition ($R^2 \leq 0.11$), motivating the R1 revision to drop the shape partition entirely (Section 9.4).*

7 Phase 3: Cohort Encoding and Harmonisation

7.1 Overview

Phase 3 encodes all MRI volumes from both the BraTS-MEN cross-sectional cohort and the Andalusian longitudinal cohort through the frozen encoder–SDP pipeline, producing per-subject latent vectors and per-patient temporal trajectories suitable for GP-based growth prediction.

The pipeline is fully deterministic (no dropout, model in evaluation mode) and proceeds as:

$$\mathbf{x} \in \mathbb{R}^{4\times 192^3} \xrightarrow{\text{SwinViT (frozen)}} \xrightarrow{\texttt{encoder10}} \xrightarrow{\text{GAP}} \mathbf{h} \in \mathbb{R}^{768} \xrightarrow{g_\theta \text{ (frozen SDP)}} \mathbf{z} \in \mathbb{R}^{128}. \tag{34}$$

7.2 Sliding Window Feature Extraction

For volumes larger than the 128^3 training ROI, or to ensure consistent feature extraction regardless of tumour position, we employ a sliding window approach with tumour-weighted averaging:

$$\mathbf{h} = \frac{\sum_{p=1}^{P} w_p \cdot \mathrm{GAP}\Big(\texttt{encoder10}(\mathbf{x}_p)\Big)}{\sum_{p=1}^{P} w_p}, \quad w_p = \frac{|\mathrm{mask}_p \cap \mathrm{tumour}|}{|\mathrm{patch}_p|}, \tag{35}$$

where $\mathbf{x}_p$ denotes patch p, and w_p weights each patch's contribution by its tumour content fraction. Patches with no tumour ($w_p = 0$) are excluded. This ensures that the feature vector is dominated by tumour-containing regions.

In practice, the feature extraction transforms use the full 192^3 ROI (see Section 3.4), which provides 100% tumour containment for meningiomas without requiring sliding window aggregation. The sliding window protocol is retained for robustness and for use with the Andalusian cohort where acquisition protocols may vary.

7.3 ComBat Harmonisation

Scanner-induced batch effects in multi-site MRI data can introduce spurious variance in the latent space that confounds temporal growth patterns. We assess and optionally apply ComBat harmonisation [13, 14] to the encoded latent vectors.

7.3.1 Assessment Protocol

Before applying ComBat, we assess the degree of scanner-induced shift using:

1. **MMD between cohorts:** $\mathrm{MMD}^2(\mathbf{z}_{\mathrm{BraTS}}, \mathbf{z}_{\mathrm{Andalusian}})$ with permutation test.
2. **UMAP visualisation:** Coloured by site/scanner, to visually assess clustering by acquisition rather than biology.
3. **Kolmogorov–Smirnov tests:** Per-dimension KS tests between sites.

If the MMD permutation test p-value < 0.05, ComBat harmonisation is applied. Otherwise, harmonisation is skipped to avoid unnecessary smoothing of biological signal.

7.3.2 ComBat Formulation

ComBat models each latent dimension j at site s as:

$$z_{ij} = \alpha_j + \mathbf{x}_i^\top \boldsymbol{\beta}_j + \gamma_{sj} + \delta_{sj}\, \epsilon_{ij}, \tag{36}$$

where α_j is the overall mean, $\mathbf{x}_i$ are biological covariates (age, sex if available), γ_{sj} and δ_{sj} are site-specific location and scale effects, and ϵ_{ij} is the residual. Empirical Bayes

estimation of γ and δ provides stable harmonisation even with small site sample sizes. The harmonised value is:

$$z_{ij}^* = \frac{z_{ij} - \hat{\alpha}_j - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_j - \hat{\gamma}_{sj}}{\hat{\delta}_{sj}} + \hat{\alpha}_j + \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_j. \tag{37}$$

7.3.3 Scanner Consistency Verification

A critical prerequisite is verifying scanner consistency within each patient's longitudinal series. If a patient was scanned on different scanners across visits, standard ComBat (which assumes one site label per observation) may distort temporal trends. In this case, LongComBat [48], which explicitly models within-subject scanner changes in a mixed-effects framework, should be used instead.

Remark 6. *ComBat's temporal preservation property requires careful verification. For a patient with timepoints t_1, t_2 scanned on the same scanner, the temporal difference $\Delta\mathbf{z} = \mathbf{z}(t_2) - \mathbf{z}(t_1)$ should be preserved up to the site-specific scale factor $\hat{\delta}_s$. This is tested explicitly in the verification protocol (Section 9).*

7.4 Trajectory Construction

For each patient i in the longitudinal cohort, we construct a temporal trajectory by:

1. Encoding all n_i MRI studies through the frozen pipeline, yielding $\{\mathbf{z}_i(t_{ij})\}_{j=1}^{n_i}$.
2. Computing time offsets t_{ij} in months from the patient's first scan: $t_{ij} = (\text{date}_{ij} - \text{date}_{i1})/30.44$.
3. Verifying temporal ordering: $t_{i1} < t_{i2} < \cdots < t_{in_i}$.
4. Extracting the partition subvectors: $\mathbf{z}_{\text{vol}}(t_{ij}) = \mathbf{z}_i(t_{ij})[0:24]$, etc.

The output is a set of per-patient trajectories:

$$\mathcal{T}_i = \Big\{\Big(\mathbf{z}_i(t_{ij}),\ t_{ij}\Big)\Big\}_{j=1}^{n_i}, \quad i = 1, \ldots, N. \tag{38}$$

7.5 Cohort Distribution Comparison

Figure placeholder: Cohort distribution comparison.
UMAP of BraTS-MEN (blue) vs. Andalusian (red) latent vectors.
Left: before ComBat. Right: after ComBat (if applied).

Figure 3: UMAP comparison of BraTS-MEN ($n = 1{,}000$) and Andalusian ($n = 137$) latent representations. The degree of overlap indicates the compatibility of the two cohorts in the learned latent space.

Figure placeholder: Patient trajectories in latent space.
2D UMAP with temporal arrows for 5–10 patients with $n_i \geq 3$.

Figure 4: Latent space trajectories for selected patients with ≥ 3 timepoints. Arrows indicate temporal progression. Smooth, directed trajectories suggest that the latent space captures meaningful temporal dynamics.

8 Phase 4: GP-Based Growth Prediction

8.1 Overview

Phase 4 fits and evaluates three growth prediction models of increasing complexity on the SDP latent trajectories from the Andalusian cohort. All models produce volume

predictions via decoding through the frozen semantic head π_{vol}.

8.1.1 Why Not a Neural ODE?

The original pipeline design used a Neural ODE [49, 50] for temporal modelling. This was abandoned due to catastrophic overparameterisation: with 33 patients (100 observations, 112 forward pairs), the Neural ODE's ∼3,100+ parameters produce a parameter-to-observation ratio of approximately 27.7. This makes reliable hyperparameter estimation impossible and yields degenerate overfitting. The full pivot rationale is documented in [51].

The GP hierarchy provides:

- **Appropriate complexity:** 6–95 parameters (vs. 3,100+).
- **Closed-form inference:** No iterative ODE solving; predictive distributions are analytic (Equations (8) and (9)).
- **Calibrated uncertainty:** GP posterior naturally provides uncertainty that increases with extrapolation distance.
- **Principled handling of heterogeneous observation counts:** Patients with $n_i = 2$ produce wide uncertainty bands that shrink toward the population mean, while patients with $n_i = 6$ receive tight, data-driven predictions.

8.2 Model A: Linear Mixed-Effects Model (LME)

The simplest model treats each latent volume dimension independently with a random-intercept, random-slope linear mixed-effects model. For patient i, dimension $d \in \{1, \ldots, 24\}$, at time t_{ij} (months from first scan):

$$z_i^{(d)}(t_{ij}) = \underbrace{(\beta_0^{(d)} + b_{0i}^{(d)})}_{\text{intercept}} + \underbrace{(\beta_1^{(d)} + b_{1i}^{(d)})}_{\text{slope}} \cdot t_{ij} + \epsilon_{ij}^{(d)}, \tag{39}$$

where $(\beta_0^{(d)}, \beta_1^{(d)})$ are the fixed effects (population intercept and slope), $(b_{0i}^{(d)}, b_{1i}^{(d)}) \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega}^{(d)})$ are patient-specific random effects, and $\epsilon_{ij}^{(d)} \sim \mathcal{N}(0, \sigma^{(d)2})$ is the residual noise.

8.2.1 Parameter Count

Per dimension: 2 fixed effects (β_0, β_1), 3 variance parameters (2 diagonal + 1 off-diagonal of $\mathbf{\Omega}$), 1 residual variance (σ^2) = 6 parameters. For 24 dimensions: $6 \times 24 = 144$ total parameters, estimated from all 100 observations via Restricted Maximum Likelihood (REML) using `statsmodels.MixedLM`.

8.2.2 Single-Patient Inference

For a held-out patient, the Best Linear Unbiased Predictor (BLUP) [52] provides patient-specific predictions with automatic shrinkage: patients with few observations ($n_i = 2$) receive predictions closer to the population mean, while patients with many observations ($n_i = 6$) receive data-driven predictions.

8.3 Model B: Hierarchical Gaussian Process (H-GP)

The H-GP extends Model A by replacing the linear temporal model with a GP, enabling nonlinear growth trajectories while retaining the population mean from the LME as the GP prior:

$$z_i^{(d)}(t) \sim \mathcal{GP}\Big(m^{(d)}(t),\ k^{(d)}(t,t')\Big), \tag{40}$$

where the mean function is the LME population trend:

$$m^{(d)}(t) = \hat{\beta}_0^{(d)} + \hat{\beta}_1^{(d)} \cdot t, \tag{41}$$

and the temporal kernel is Matérn-5/2 (Equation (11)):

$$k^{(d)}(t,t') = k_{5/2}(|t-t'|;\ \sigma_f^{(d)}, \ell^{(d)}) + \sigma_n^{(d)2}\,\delta(t,t'). \tag{42}$$

8.3.1 Hierarchical Hyperparameter Sharing

With $n_i \in [2,6]$ observations per patient, per-patient kernel fitting is ill-conditioned. We adopt empirical Bayes: shared kernel hyperparameters $\boldsymbol{\theta}^{(d)} = (\sigma_f^{(d)}, \ell^{(d)}, \sigma_n^{(d)})$ are estimated by maximising the pooled log-marginal likelihood across all 33 patients:

$$\hat{\boldsymbol{\theta}}^{(d)} = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{N} \log p\Big(\mathbf{z}_i^{(d)} \mid \mathbf{t}_i, \boldsymbol{\theta}\Big). \tag{43}$$

Given shared hyperparameters, each patient's predictive distribution is computed by conditioning on their individual observations via Equations (8) and (9).

8.3.2 Parameter Count

3 hyperparameters per dimension $\times$ 24 dimensions = 72 total. Optimised via L-BFGS-B with 5 random restarts.

8.4 Model C: Partition-Aware Multi-Output GP (PA-MOGP)

The PA-MOGP is the most complex model, jointly modelling volume, location, and shape dynamics with partition-specific temporal kernels and cross-partition coupling. It operates

on the active latent subspace:

$$\tilde{\mathbf{z}} = [\mathbf{z}_{\text{vol}}^{24} \mid \mathbf{z}_{\text{loc}}^{8} \mid \mathbf{z}_{\text{shape}}^{12}] \in \mathbb{R}^{44}, \tag{44}$$

excluding the residual partition (84 unsupervised dimensions that would cause overfitting).

8.4.1 Partition-Specific Temporal Kernels

Each partition uses a kernel matched to its expected temporal dynamics:

Table 15: Partition-specific temporal kernels in the PA-MOGP.

Partition	**Kernel**	**Rationale**
$\mathbf{z}_{\text{vol}}$ (24 dims)	Matérn-5/2	Smooth, twice-differentiable growth
$\mathbf{z}_{\text{loc}}$ (8 dims)	Squared Exponential	Very slow, infinitely smooth centroid drift
$\mathbf{z}_{\text{shape}}$ (12 dims)	Matérn-3/2	Once-differentiable; shape can change abruptly

8.4.2 Composite Kernel with Cross-Partition Coupling

The PA-MOGP uses an Intrinsic Coregionalisation Model (Equation (12)) with an additional rank-1 coupling term:

$$\mathbf{K}(t,t') = \underbrace{\mathbf{B}_{\text{vol}}^{\text{diag}} \otimes k_{\text{vol}} + \mathbf{B}_{\text{loc}}^{\text{diag}} \otimes k_{\text{loc}} + \mathbf{B}_{\text{shape}}^{\text{diag}} \otimes k_{\text{shape}}}_{\text{Within-partition dynamics}} + \underbrace{\mathbf{B}_{\text{cross}} \otimes k_{\text{vol}}}_{\text{Cross-partition coupling}} + \sigma_n^2 \mathbf{I}_{44}\, \delta(t,t'), \tag{45}$$

where $\mathbf{B}_{\text{cross}} = \mathbf{w}\mathbf{w}^\top$ with $\mathbf{w} \in \mathbb{R}^{44}$ (rank-1 coupling). The coupling encodes the mechanistic hypothesis that volume growth is the primary driver and shape/location changes are secondary consequences, operating on the same timescale as volume dynamics.

8.4.3 Parameter Count

Table 16: PA-MOGP parameter breakdown.

Component	**Parameters**
Vol kernel $(\sigma_f, \ell, \sigma_n)$	3
Loc kernel $(\sigma_f, \ell, \sigma_n)$	3
Shape kernel $(\sigma_f, \ell, \sigma_n)$	3
Vol diagonal coregionalisation (24 dims)	24
Loc diagonal coregionalisation (8 dims)	8
Shape diagonal coregionalisation (12 dims)	12
Cross-partition coupling $\mathbf{w}$	44
Noise variance	1
Total	**98**

Hyperparameters are estimated via pooled multivariate marginal likelihood (4,400 observation-dimension pairs) using L-BFGS-B with 5 random restarts.

Remark 7 (R1 Revision)**.** *Following the methodology revision, the PA-MOGP simplifies to a standard MOGP operating on* $\mathbf{z}_{vol} \in \mathbb{R}^{32}$ *only (with a single Matérn-5/2 kernel), since the shape partition is dropped and location becomes a static GP covariate. The cross-partition coupling term is removed, and the coregionalisation matrix* $\mathbf{B} = \mathbf{W}\mathbf{W}^\top + \mathrm{diag}(\boldsymbol{\kappa})$ *captures inter-dimension correlations within the volume partition.*

8.5 Leave-One-Patient-Out Cross-Validation

With only 33 patients, k-fold CV leaves too few patients per fold for stable GP hyperparameter estimation. LOPO-CV uses maximum training data per fold (32 patients, ∼97 observations):

Algorithm 1 LOPO-CV Evaluation Protocol

```
for i = 1 to N = 33 do
    D_train ← {T_j : j ≠ i} {32 patients}
    D_test ← {T_i} {held-out patient}
    Fit model on D_train
    Predict at patient i's observation times
    Decode predictions through π_vol
    Record per-patient metrics
end for
Aggregate metrics across 33 folds
```

For each held-out patient i, predictions are evaluated at the patient's observed time-points using the model trained on the remaining 32 patients. The LME model is refitted

from scratch in each fold. For H-GP and PA-MOGP, shared hyperparameters are re-optimised per fold (excluding patient i's data from the pooled marginal likelihood).

8.6 Volume Decoding

Predicted latent vectors are decoded to physical volumes through the frozen SDP semantic head:

$$\hat{v} = \pi_{\text{vol}}(\hat{\mathbf{z}_{\text{vol}}}) = \mathbf{W}_v \hat{\mathbf{z}_{\text{vol}}} + \mathbf{b}_v, \quad \hat{v} \in \mathbb{R}^4, \tag{46}$$

where $\hat{v}$ represents log-volumes in normalised space. De-normalisation applies the stored training statistics:

$$V_{\text{pred}} = \exp\Big(\hat{v} \cdot \sigma_{\text{vol}} + \mu_{\text{vol}}\Big) - 1, \tag{47}$$

yielding predicted volumes in mm^3.

8.6.1 Uncertainty Propagation

For GP models that provide predictive covariance $\mathbf{\Sigma}_{\hat{\mathbf{z}_{\text{vol}}}}$, the decoded volume uncertainty is propagated via the linear map:

$$\mathbf{\Sigma}_{\hat{v}} = \mathbf{W}_v \, \mathbf{\Sigma}_{\hat{\mathbf{z}_{\text{vol}}}} \, \mathbf{W}_v^\top. \tag{48}$$

In log-volume space, this provides Gaussian uncertainty bands. Transformation to physical volume space requires the delta method or Monte Carlo sampling for non-Gaussian propagation through the exponential.

8.7 Population Mean Function

The GP prior mean function is set to the LME population trend (Equation (41)). This creates a natural model nesting:

$$\mathcal{GP} \text{ with linear mean} \supset \text{LME}, \tag{49}$$

ensuring that for patients with $n_i = 2$, the GP predictions revert to the well-estimated population trend rather than the zero function. Alternative mean functions (zero, Gompertz) are evaluated as ablation A8 (Section 9.2).

8.8 Failure Recovery Protocol

If GP hyperparameter fitting diverges or produces degenerate results, the following recovery steps are applied in order:

1. Add diagonal jitter to the kernel matrix: $\mathbf{K} \leftarrow \mathbf{K} + 10^{-6}\,\mathbf{I}$.

2. Constrain length-scale bounds: $\ell \in [1, 120]$ months.

3. Increase L-BFGS-B random restarts from 5 to 10.

4. Fall back to a simpler kernel (Matérn-5/2 $\rightarrow$ SE).

5. For PA-MOGP: disable cross-partition coupling (reduces to independent per-partition GPs).

6. If all fail: report per-dimension diagnostics and halt.

9 Evaluation

9.1 Quality Targets

The pipeline defines quantitative quality targets at each phase, with minimum thresholds that block further progress and aspirational targets for optimal performance.

Table 17: Phase 1 quality targets (encoder adaptation).

Metric	**Target**	**Minimum**	**Status**
Dice (WT) on `lora_val`	≥ 0.85	≥ 0.80	*[to fill]*
Dice improvement over frozen BSF	≥ 0.05	≥ 0.02	*[to fill]*
Linear probe Vol R^2 (adapted)	≥ 0.50	≥ 0.30	*[to fill]*

Table 18: Phase 2 quality targets (SDP).

Metric	**Target**	**Minimum**	**Status**
Vol R^2 on validation	≥ 0.90	≥ 0.80	*[to fill]*
Loc R^2 on validation	≥ 0.95	≥ 0.85	*[to fill]*
Shape R^2 on validation	≥ 0.40	≥ 0.25	*[to fill]*
Max cross-partition corr	< 0.20	< 0.30	*[to fill]*
Dims with var > 0.5	$\geq 95\%$	$\geq 85\%$	*[to fill]*
$\mathrm{dCor}(\mathrm{vol}, \mathrm{loc})$	< 0.10	< 0.20	*[to fill]*

Table 19: Phase 4 quality targets (growth prediction).

Metric	**Target**	**Minimum**	**Status**
Vol prediction R^2 (LOPO, best model)	≥ 0.70	≥ 0.50	*[to fill]*
Calibration (95% CI coverage, GP models)	0.90–0.98	≥ 0.80	*[to fill]*
Per-patient trajectory r ($n_i \geq 3$)	≥ 0.80	≥ 0.60	*[to fill]*
PA-MOGP $R^2 >$ H-GP R^2	> 0	—	*[to fill]*

9.2 Ablation Study Design

Ten ablation experiments systematically evaluate design decisions across all pipeline phases:

Table 20: Full ablation matrix (A1–A10).

ID	Variable	Conditions	Primary metric
A1	LoRA rank	$\{2, 4, 8, 16, 32\}$	Dice, probe R^2
A2	LoRA vs. DoRA	$\{\text{LoRA}, \text{DoRA}\}$ at $r = 8$	Dice, probe R^2
A3	Aux semantic heads	{with, without}	Phase 2 R^2
A4	SDP dimension	$\{64, 128, 256\}$	R^2, dCor
A5	VICReg + dCor	{full, no cov, no dCor, no both}	Cross-partition corr
A6	Growth model	$\{\text{LME}, \text{H-GP}, \text{PA-MOGP}\}$	Vol R^2 (LOPO)
A7	ComBat effect	{with, without}	Vol R^2 (LOPO)
A8	GP mean function	{zero, linear, Gompertz}	Vol R^2 (LOPO)
A9	GP kernel	{M-3/2, M-5/2, SE}	Vol R^2 (LOPO)
A10	Cross-partition coupling	{with, without}	Vol R^2, calibration

9.2.1 A6: Model Comparison

The head-to-head comparison of the three GP models is the central evaluation of the pipeline. The comparison uses LOPO-CV and reports:

Vol R^2: Coefficient of determination between predicted and observed whole-tumour volume across all held-out observations.

Vol MAE: Mean absolute error in physical volume (mm^3).

Latent MSE: Mean squared error in the latent volume partition $\mathbf{z}_{\text{vol}}$ (before decoding).

Calibration: Fraction of true observations within the predicted 95% confidence interval (target: 0.90–0.98).

Per-patient r: Pearson correlation between predicted and observed volume trajectories for patients with $n_i \geq 3$.

Table 21: Model comparison results (LOPO-CV, 33 folds).

Model	Params	Vol R^2	Vol MAE	Calibration	Per-patient r
LME	144	*[to fill]*	*[to fill]*	—	*[to fill]*
H-GP	72	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
PA-MOGP	98	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*

9.3 Publication Figures

The evaluation produces 13 publication-quality figures documenting each pipeline phase. Key figures include:

Figure placeholder: Volume prediction scatter.
Predicted vs. actual ΔV for all LOPO-CV test pairs,
coloured by model (LME: blue, H-GP: orange, PA-MOGP: green).

Figure 5: Predicted vs. actual volume change (ΔV, mm^3) across all LOPO-CV held-out observation pairs, coloured by model. The diagonal represents perfect prediction. Dispersion around the diagonal quantifies prediction error.

Figure placeholder: Trajectory prediction with uncertainty.
3–4 example patients showing predicted trajectory
(mean $\pm$ 95% CI) overlaid on actual observations.

Figure 6: Growth trajectory predictions for selected patients. Solid lines show GP posterior mean; shaded regions show 95% confidence intervals. Dots are actual observations. The GP uncertainty naturally widens with extrapolation distance.

Figure placeholder: Model comparison bar chart.
R^2, MAE, and calibration for each model.

Figure 7: Head-to-head model comparison across three metrics. Error bars represent bootstrap 95% confidence intervals computed over the 33 LOPO-CV folds.

Figure placeholder: Disentanglement matrix.
Cross-partition correlation heatmap from SDP.

Figure 8: Cross-partition correlation heatmap after SDP training. Each cell shows the maximum absolute Pearson correlation between dimensions in the corresponding partition pair. Diagonal blocks (within-partition) show internal structure; off-diagonal blocks should approach zero for successful disentanglement.

Figure placeholder: Learned GP length-scales per partition. Revealing characteristic timescales of volume/location/shape dynamics.

Figure 9: Learned GP kernel length-scales (ℓ) for the PA-MOGP model. Volume dynamics have the longest characteristic timescale (slow growth), while shape dynamics are faster (shorter ℓ). Location length-scales are expected to be very long (near-static centroid).

9.4 R1 Methodology Revision

Empirical findings from Phases 1–2 motivated three principled revisions to the pipeline architecture, collectively designated the R1 revision.

9.4.1 Shape Partition Failure

Across all experimental conditions, the shape partition achieved $R^2 \leq 0.11$ for linear probes and approximately zero for RBF probes. Cross-domain shape transfer was uniformly catastrophic ($R^2 = -2.00$). This failure is biologically grounded: gliomas are infiltrative and irregular, while meningiomas are well-circumscribed and dural-based. Shape semantics differ qualitatively between the two tumour types, making a shared shape representation a category error.

Table 22: Shape probe R^2 across experimental conditions.

Condition	**Linear R^2**	**RBF R^2**	**Cross-domain**
Frozen baseline	0.08	≈ 0	-2.00
MEN LoRA r=8	0.11	≈ 0	-2.00
Dual LoRA r=8	-0.06	≈ 0.15	-2.00

Decision: Drop the shape partition entirely from the SDP.

9.4.2 Location: Temporally Static

Meningiomas are extra-axial, dural-attached tumours that do not migrate. The centroid coordinates at t_0 are essentially identical at subsequent timepoints (modulo asymmetric growth). Modelling location as a time-varying GP dimension fits a constant function, which is statistically vacuous and wastes degrees of freedom. Within-domain location R^2 is reasonable (0.35 MEN, 0.68 GLI), confirming the information is present but does not need temporal modelling.

Decision: Relocate location from temporal latent dimension to static GP covariate, entering the mean function as $m_i(t) = \beta_0 + \beta_1 t + \boldsymbol{\gamma}^\top \mathbf{c}_i$, where $\mathbf{c}_i$ is the centroid from patient i's initial scan.

9.4.3 Volume Target Simplification

The 4-dimensional volume target $[\log V_{\text{total}}, \log V_{\text{NCR}}, \log V_{\text{ED}}, \log V_{\text{ET}}]$ is problematic because: (i) label semantics differ across tumour types (GLI oedema $\neq$ MEN oedema); (ii) sub-compartment segmentation quality is poor (TC Dice 0.40 on MEN); (iii) whole-tumour volume is the only clinically relevant growth endpoint for meningiomas.

Decision: Reduce to a single target $v^* = \log(V_{\text{WT}} + 1) \in \mathbb{R}^1$.

9.4.4 Revised Architecture Summary

$$\text{R1:} \quad \mathbf{z} \in \mathbb{R}^{128} = [\underbrace{\mathbf{z}_{\text{vol}} \in \mathbb{R}^{32}}_{\text{supervised}} \mid \underbrace{\mathbf{z}_{\text{res}} \in \mathbb{R}^{96}}_{\text{unsupervised}}], \quad \pi_{\text{vol}} : \mathbb{R}^{32} \to \mathbb{R}^1. \tag{50}$$

The PA-MOGP simplifies to a standard MOGP on $\mathbb{R}^{32}$ with a single Matérn-5/2 kernel and ICM coregionalisation matrix $\mathbf{B} = \mathbf{W}\mathbf{W}^\top + \text{diag}(\boldsymbol{\kappa})$.

9.5 LoRA Marginality

A notable finding is the marginal benefit of LoRA adaptation for downstream feature quality. The frozen BrainSegFounder baseline achieves $R^2 = 0.794$ for volume probing, while the best LoRA condition adds only $\Delta R^2 = +0.016$ (within noise). This suggests that BrainSegFounder's SSL pretraining on 41,400+ subjects already produces features with strong meningioma volume decodability.

Several hypotheses explain this observation:

1. **SSL pretraining captures domain-general features.** The UK Biobank pretraining stage exposes the encoder to massive anatomical diversity, producing features that generalise across brain pathologies.

2. **LoRA's low-rank constraint limits adaptation.** With $r = 8$ and $< 0.32\%$ trainable parameters, LoRA may not have sufficient capacity to substantially restructure the 768-dimensional feature space.

3. **Segmentation is a weak proxy for semantic quality.** Improving segmentation Dice does not necessarily improve the linear decodability of aggregate semantic properties (volume, location).

This finding has important practical implications: for meningioma growth prediction, the frozen BrainSegFounder encoder may be a sufficient starting point, with the SDP and GP models providing the primary value.

10 Conclusion

10.1 Summary of Contributions

This thesis has presented a four-phase pipeline for meningioma growth forecasting from longitudinal multi-modal 3D MRI. The key contributions and findings are:

1. **Foundation model adaptation.** We demonstrated that BrainSegFounder, pretrained on 41,400+ subjects for glioma segmentation, can be adapted to the meningioma domain via LoRA with ∼197K trainable parameters ($< 0.32\%$ of total). However, the marginal improvement in downstream feature quality ($\Delta R^2 = +0.016$) suggests that SSL pretraining already provides strong cross-domain features, and the primary benefit of LoRA is in segmentation quality rather than representation structure.

2. **Supervised disentangled projection.** We designed a non-generative approach to structured latent space construction, theoretically grounded in the impossibility of unsupervised disentanglement [15] and validated against the failure of VAE-based alternatives. The combination of semantic regression, VICReg regularisation, and distance correlation provides a principled and computationally efficient framework.

3. **Empirically-driven methodology revision.** Rigorous evaluation revealed that shape disentanglement fails ($R^2 \leq 0.11$), meningioma location is temporally static, and sub-compartment volume targets are label-dependent and clinically irrelevant. These findings motivated a principled simplification from a $3 + 1$ partition SDP with PA-MOGP to a $1 + 1$ partition SDP with standard MOGP.

4. **GP hierarchy for small-sample temporal modelling.** The three-model hierarchy (LME → H-GP → PA-MOGP) replaces the originally proposed Neural ODE, which was catastrophically overparameterised for 33 patients (∼3,100 parameters vs. 112 observation pairs). The GP framework provides closed-form inference, calibrated uncertainty, and principled handling of heterogeneous observation counts.

5. **Negative results on dual-domain transfer.** Cross-domain GP probes yield $R^2 \approx -0.01$, demonstrating that domain gap reduction (MMD^2 decrease of 68.6%) does not

guarantee semantic compatibility. Dual-domain training compresses the effective rank by 46.6% while degrading MEN segmentation Dice from 0.64 to 0.36.

10.2 Limitations

10.2.1 Cohort Size

The Andalusian longitudinal cohort (33 patients, 100 observations) is the primary bottleneck. With 57.6% of patients having only 2 studies, the per-patient growth trajectories are sparse. While the GP framework handles this gracefully via automatic shrinkage toward the population mean, the resulting predictions for 2-observation patients are dominated by the population trend rather than individual-specific dynamics.

10.2.2 Temporal Sparsity

The irregular and sparse temporal sampling (2–6 observations per patient over months to years) limits the detection of non-monotonic growth patterns (e.g., saltatory growth, temporary regression). The Matérn-5/2 kernel imposes a smoothness prior that may be inappropriate for tumours with abrupt growth regime changes.

10.2.3 LoRA Marginality

The finding that LoRA adaptation provides marginal downstream benefit ($\Delta R^2 = +0.016$) raises the question of whether the Phase 1 adaptation step is necessary at all for this pipeline. The frozen BrainSegFounder encoder may suffice, with the SDP and GP models providing the primary modelling value. However, LoRA does improve segmentation quality, which affects the accuracy of the semantic feature extraction used to train the SDP.

10.2.4 Single-Site Longitudinal Data

The Andalusian cohort is collected from hospitals in a single region, which may limit generalisability to other populations, scanner manufacturers, and acquisition protocols. ComBat harmonisation addresses scanner batch effects but cannot correct for biological population differences.

10.2.5 Absence of Clinical Validation

The pipeline has not been validated against clinical outcomes (intervention decisions, actual tumour behaviour at longer follow-up). The volume prediction R^2 and calibration metrics are necessary but not sufficient for clinical utility.

10.3 Future Work

10.3.1 Larger Longitudinal Cohorts

The most impactful improvement would be access to larger longitudinal meningioma cohorts. Multi-centre collaborations could provide 100–500 patients, which would enable: (i) proper training/validation/test splits for GP models; (ii) exploration of higher-capacity models (deep kernel GPs, neural process families); (iii) subgroup analysis by tumour grade, location, and patient demographics.

10.3.2 Self-Supervised Temporal Pretraining

Following the JEPA paradigm [36], the SDP and GP models could be replaced by a self-supervised temporal prediction objective: given the latent vector at t_0, predict the latent vector at t_1 in representation space. This would require more longitudinal data but could learn temporal dynamics end-to-end rather than relying on the two-stage SDP + GP approach.

10.3.3 Clinical Decision Support Integration

The GP posterior's calibrated uncertainty is directly suitable for clinical decision support: the probability that a tumour will exceed a volume threshold V_{critical} within a specified time horizon can be computed analytically from the Gaussian predictive distribution. Integrating this into a clinical workflow would require prospective validation and regulatory approval.

10.3.4 Multi-Task Growth Prediction

Rather than predicting volume alone, the pipeline could be extended to jointly predict clinical endpoints (e.g., symptom onset, need for surgery) from the latent trajectories. This would require outcome labels that are typically available in retrospective clinical databases.

10.3.5 Foundation Model Scaling

BrainSegFounder is a relatively small foundation model (BSF-Tiny, 62M parameters). Larger variants or more recent foundation models pretrained on broader datasets may provide stronger initial features, potentially eliminating the need for LoRA adaptation entirely and improving the SDP's semantic decodability.

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